LCA Module v3 — Merge Specification Document

Version: 3.0.0

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1. Basis and Rationale for the Merge

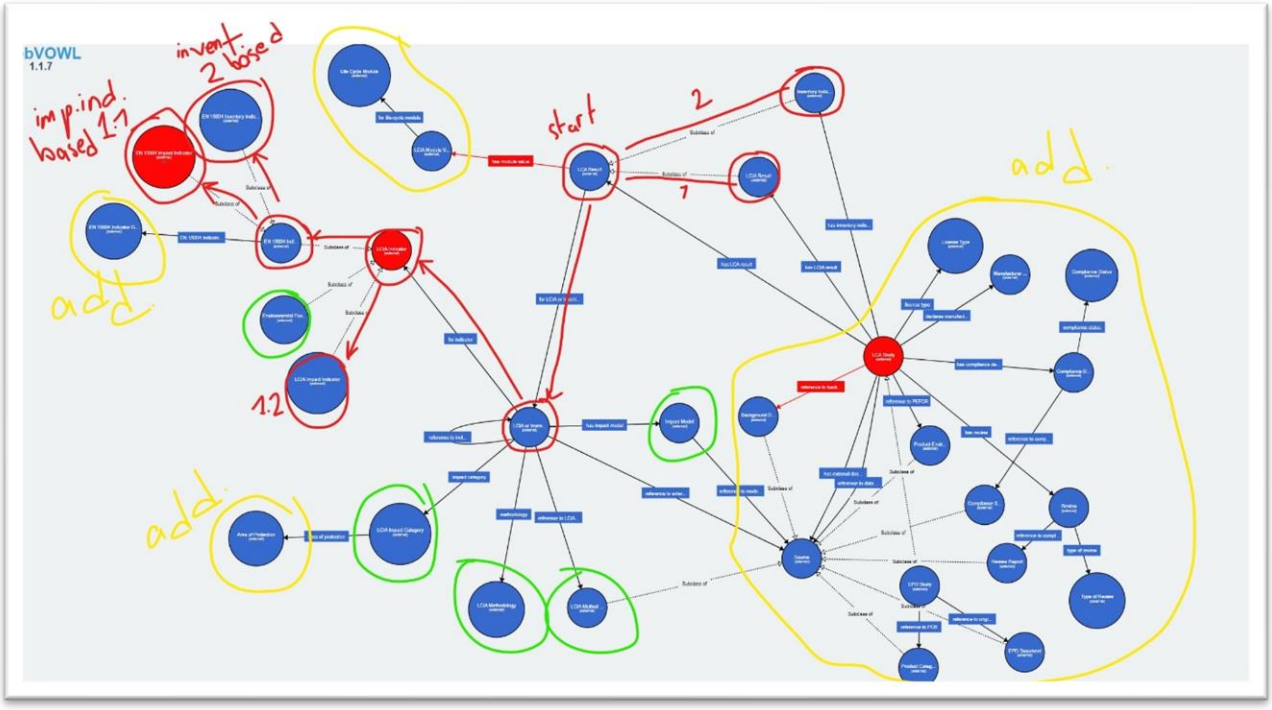
The LCA module v3 results from the integration of two independently developed v2.0:

- **Maki v2.0** — developed by Maki, structured around ILCD+EPD with a lean manner integrating EN 15804+A2 indicator individuals, lifecycle module value observations.
- **CEA v2.0** — developed by CEA List, structured around ESPR only as the regulatory alignment with rich semantics along the class/property.

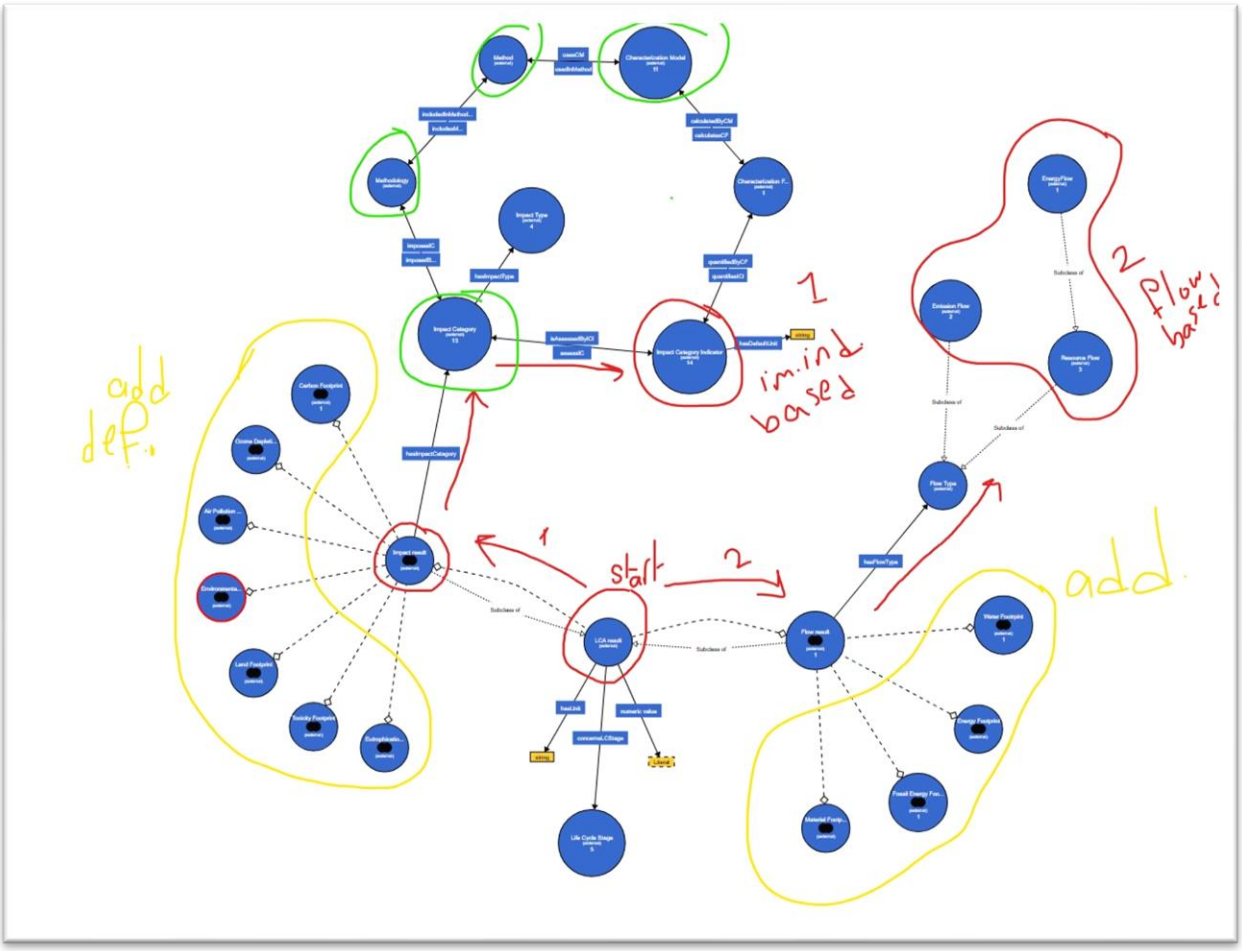
Following a comparative analysis of both ontologies — documented in the annotated diagrams attached to this specification (color-coded: red = structural similarity - a shared backbone, green = same concepts - modeled differently, yellow = additions of each,) — **Maki v2.0 was selected as the base for v3, with CEA's structural modifications to be incorporated within**. This decision rests on two grounds:

1. Maki's lean scope matches the DPP mandatory and non-mandatory regulations (ESPR, ILCD-base, ILCD+EPD extension) as well as following the EN15804+A2 and EF 3.1 controlled vocabulary of named individuals (19 EN 15804+A2 impact indicators, 18 inventory indicators, 16 EF 3.1 indicators) providing thus a **standards-grounded base**.
2. . CEA's structual formalization of classes/properties (e.g., indicator-centric result architecture (`LCAResult → quantifiesIndicator → LCIAIndicator`)) is semantically more correct than Maki's list-based declaration of classes/properties, having the ability thus to add **richer and cleaner semantics** to an extant base.

The changes and additions constituting v3 are documented in full in the sections below.



LCA_maki_v2.0.0



LCA_cea_v2.0.0

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2. Minor ontology Metadata Changes

2.1 Description Block

The following disclaimer block has been **removed** from the ontology's `dcterms:description`:

> *"DISCLAIMER: The scope is deliberately lean…" (full text removed — the scope statement is now implicit in the v3 structure itself and documented here instead of in the ontology header.)*

2.2 Namespace Declarations Added

The following namespace prefixes have been added to the ontology header:

```
xmlns:spdx="http://spdx.org/rdf/terms#"
xmlns:foaf="http://xmlns.com/foaf/0.1/"
xmlns:prov="http://www.w3.org/ns/prov#"
```

2.3 Versioning

- `owl:versionInfo`: "3.0.0"
- `owl:versionIRI`: https://w3id.org/eudpp/LCA/3.0.0
- `dcterms:creator`: Fatima Danash; Marc-Andree Wolf

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3. Minor class Hierarchy Changes

3.1 Renames

Previous name	v3 name	Reason
`LCIAModuleValue`	`LCAModuleValue`	Not specific to LCIA — also covers inventory module observations
`LCIAIndicator`	`LCAIndicator`	Same
`LCIAMethdology`	`LCAMethodology`	Same
`LCIAOrInventoryMethod`	`LCA Method`	Simpler; the qualifier was redundant given the ontology scope
`EnvironmentalFootprintScore`	`EnvironemntalFootprintScoreResult`	It is clearly a result

ALL comments have been respectively clarified.

3.2 Removed Classes

**`Unit` class and `hasUnit` property`** — removed. Unit representation is already handled in the P_DPP module via the SI Digital Framework `MeasurementUnit` class and the `hasMeasurementUnit` object property (linking `QuantitativeProperty` to `MeasurementUnit`). A connector-level link between `LCAResult` and `QuantitativeProperty` will allow LCA results to inherit `hasMeasurementUnit` from P_DPP without redefining it here.

3.3 External Alignments Added

The following `rdfs:subClassOf` and `skos:broadMatch` alignments have been added to improve interoperability with standard vocabularies:

Class	Alignment	Target
`Source`	`rdfs:subClassOf`	`prov:Entity`
`ComplianceDeclaration`	`rdfs:subClassOf`	`prov:Entity`
`EPDDocument`	`skos:broadMatch`	`foaf:Document`
`LCIAMethodReport`	`skos:broadMatch`	`foaf:Document`
`PCR`	`skos:broadMatch`	`foaf:Document`
`PEFCR`	`skos:broadMatch`	`foaf:Document`
`ReviewReport`	`skos:broadMatch`	`foaf:Document`
`LicenseType`	`rdfs:subClassOf`	`spdx:License`
`Review`	`rdfs:subClassOf`	`prov:Activity`
`TypeOfReview`	`rdfs:subClassOf`	`skos:Concept`
`ComplianceStatus`	`rdfs:subClassOf`	`skos:Concept`

4. Major class Hierarchy Additions

The following classes replace P_DPP placeholder classes, which will be removed from P_DPP following Tarmo's proposal. Cross-module `rdfs:subClassOf QuantitativeProperty` declarations for all these classes are deferred to the Connector module.

4.1 CarbonFootprint

A specialisation of `LCIAResult` representing the carbon footprint of a product system — the sum of greenhouse gas emissions and removals expressed in kg CO_2 equivalent, computed over a 100-year time horizon (ESPR Art. 2(25)). Instances quantify either the EN 15804+A2 GWP total indicator (`ind_GWPtotal`, an `EN15804ImpactIndicator`) or the EF 3.1 Climate Change indicator (`ind_EF_climateChange`, an `LCIAImpactIndicator`), depending on the reporting framework of the EPD or LCA study.

OWL restriction (SubClass Of):

```
( quantifiesIndicator value ind_GWPtotal ) OR ( quantifiesIndicator value ind_EF_climateChange )
```

Connector module action: `CarbonFootprint rdfs:subClassOf QuantitativeProperty`

4.2 MaterialFootprint

A specialisation of `InventoryIndicatorResult` representing the material footprint of a product system — the total amount of raw materials extracted to meet final consumption demands (ESPR Art. 2(26)). Modelled as an inventory-level result (not a characterised LCIA result) because it aggregates raw material extraction flows without applying characterisation factors.

No `quantifiesIndicator` restriction is imposed, as material footprint aggregates across multiple raw material flow indicators in the EN 15804+A2 inventory (PERE, PENRE, SM, RSF, NRSF, etc.).

Connector module action: `MaterialFootprint rdfs:subClassOf QuantitativeProperty`

4.3 EnvironmentalFootprint (not added as a class)

ESPR Art. 2(24) defines environmental footprint as “a quantification of the environmental impacts resulting from a product throughout its life cycle, whether in relation to a single environmental impact category or an aggregated set of impact categories based on the Product Environmental Footprint method.”

This definition is intentionally broad — it covers both single-category results and fully aggregated scores. In the LCA module, `eudpp:LCAResult` already serves as the technical umbrella for all quantified LCA results. However, not every `LCAResult` constitutes an environmental footprint in the ESPR sense — `InventoryIndicatorResult` instances quantify inventory flows, not characterised impacts.

For this reason, `EnvironmentalFootprint` is **not introduced as a separate OWL class**. Instead, a `skos:broadMatch` annotation is added to `eudpp:LCAResult` pointing to the ESPR Art. 2(24) definition IRI, expressing that `LCAResult` broadly covers the regulatory concept without being formally equivalent to it.

The following sentence is added to the `rdfs:comment` of `eudpp:LCAResult`:

> *"In the regulatory sense of ESPR Art. 2(24), characterised subclasses of LCAResult (LCIAResult and EFSingleScoreResult) constitute environmental footprints — quantifications of environmental impacts of a product throughout its life cycle. InventoryIndicatorResult instances, however, quantify inventory flows rather than characterised impacts, and therefore fall outside the strict ESPR definition. For this reason, LCAResult is linked to the ESPR environmental footprint concept via skos:broadMatch rather than owl:equivalentClass."*

P_DPP action: `EnvironmentalFootprint` to be deleted from P_DPP.

4.4 EFSingleScoreResult

A specialisation of `LCAResult` representing the Environmental Footprint (EF) single score — a normalized and weighted aggregate of characterised results across all 16 EF 3.1 impact categories, expressed as a single dimensionless value (points, Pt). Computed by applying the normalization and weighting factors of the PEF method (Recommendation (EU) 2021/2279).

Modelled as a **sibling of `LCIAResult`** rather than a subclass of it, because the EF single score is qualitatively different from a per-category characterised result: it aggregates across all categories through normalization and weighting, steps that go beyond characterisation proper.

OWL restriction (SubClass Of):

```
quantifiesIndicator exactly 1 EnvironmentalFootprintScore
```

Connector module action: `EFSingleScoreResult rdfs:subClassOf QuantitativeProperty`

4.5 ResourceConsumption Subclasses

Five new classes are added as **direct subclasses of `InventoryIndicatorResult`**, replacing the placeholder `ResourceConsumption` subclasses defined in P_DPP. No intermediate `ResourceConsumption` parent class is introduced.

Class IRI	Label	Typical EN 15804+A2 indicator(s)
`WaterConsumptionResult`	Water Consumption Result	FW (net fresh water)
`EnergyConsumptionResult`	Energy Consumption Result	PERE, PERM, PERT, PENRE, PENRM, PENRT
`LandUseResult`	Land Use Result	LU (note: classified as impact indicator in EF 3.1)
`RecycledMaterialsUseResult`	Recycled Materials Use Result	SM (secondary material)
`SustainableRenewableMaterialsUseResult`	Sustainable Renewable Materials Use Result	RSF, NRSF

Each class carries an `rdfs:label` matching its IRI local name and an `rdfs:comment` describing what it measures, noting the typical — though non-binding — correspondence to the EN 15804+A2 indicators listed above. No `quantifiesIndicator` restriction is imposed on any of these subclasses, to preserve flexibility for future indicator additions or revised characterisation frameworks. The `sourceVocabulary` annotation points to the LCA module IRI (https://w3id.org/eudpp/LCA).

Note on `LandUseResult`: its placement under `InventoryIndicatorResult` reflects alignment with the P_DPP `ResourceConsumption` grouping. Reviewers should note that LU is characterised as an impact indicator under EF 3.1 in EN 15804+A2; this placement may be revisited in a future version.

Connector module: All five classes declared as `rdfs:subClassOf QuantitativeProperty`.

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5. Major object Property Changes

5.1 Deleted Properties

`forLCIAOrInventoryMethod` — deleted. This property made the `Method` instance the primary link from a result, requiring a second hop through `forIndicator` to reach the indicator. This inverts the semantic priority: an LCA result is fundamentally a quantification of an indicator, not of a method. The method is computation metadata. EN 15804+A2 and ILCD+EPD both organise result reporting rows by indicator, not by method — confirming that the indicator is the primary identity anchor of a result.

Replaced by the object property `computedByMethod`, defined between `LCIAIndicator` and `Method`, with cardinality exactly 1 (`LCIAIndicator computedByMethod exactly 1 Method`). Backward

compatibility is preserved via property chain axiom: `forLCIAOrInventoryMethod ≡ quantifiesIndicator ∘ inverse(computedByMethod)` if needed.

`forIndicator` — deleted. This property ran from `Method` to `LCIAIndicator`, making the method the subject and the indicator the object. In v3 the direction is inverted to `computedByMethod` (`LCIAIndicator → Method`), which correctly models the dependency: the indicator is the stable concept; the method is what computes it.

Replaced by object property `quantifiesIndicator` (`LCAResult → LCIAIndicator`), with cardinality exactly 1 per result instance.

`hasExternalDocumentation` — deleted (redundant with `referenceToDataSource`). Replaced by a unified property `hasExternalSource`, covering both (a) background documentation sources (project reports, safety data sheets, explanatory material) and (b) background databases or literature sources used for modelling (ecoinvent, GaBi/Sphera, Ökobaudat). The subproperty `hasReferenceToBackgroundDatabase` is retained as a specialisation for `Source` instances that are a `BackgroundDatabase`.

`hasUnit` — deleted

5.2 Renames

Previous name	v3 name
EN15804IndicatorGroup	belongsToEn15804Group
areaOfProtection	addressesAoP
referenceToIncludedMethods	includesMethod
complianceStatus	referenceToComplianceStatus
typeOfReview	hasTypeOfReview
Methodology	belongsToMethodology
referenceToDataSource	referenceToExternalSource
hasExternalDocumentation	hasExternalSource
licenseType	hasLicenseType
hasModuleValue	hasPerModuleValue
impactCategory	hasImpactCategory
referenceToExternaDocumentation(onLCIAorIventoryMethod)	referenceToExtenralSource
referenceToPCR	followsPCR
referenceToPEFCR	followsPEFCR
hasModuleValue	hasPerModuleValue

5.3 Enhanced semantics: Qualified Cardinality Restrictions on `quantifiesIndicator`

For `LCIAResult` and `InventoryIndicatorResult`, specialised range restrictions have been added via qualified cardinality axioms:

- **`LCIAResult`** is restricted to quantify exactly one indicator that is either an `EN15804ImpactIndicator` or an `LCIAImpactIndicator`. This captures the two study contexts in which a characterised impact result appears: an EPD study (EN 15804+A2 indicators) and a general LCA study (EF 3.1 / PEF indicators). The union allows both cases without forcing a choice at the class level.

- **`InventoryIndicatorResult`** is restricted to quantify exactly one `EN15804InventoryIndicator`. This reflects that inventory results are always linked to a defined inventory indicator, distinct from the characterised impact indicators used in `LCIAResult`.

The general range declaration (`quantifiesIndicator range LCIAIndicator`) on `LCAResult` is preserved and remains consistent with these restrictions, since `EN15804ImpactIndicator`, `LCIAImpactIndicator`, and `EN15804InventoryIndicator` are all subclasses of `LCIAIndicator`.

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6. Major object Property Additions

6.1 `relatesToLCIAIndicator`

Added between `EN15804ImpactIndicator` and `LCIAImpactIndicator`. Records a conceptual correspondence between the EN 15804+A2 reporting indicators and the EF 3.1 / PEF LCIA impact indicators, without implying equivalence or computational interchangeability. The property name "relates to" is deliberately the weakest possible semantic commitment — a conceptual link, not a formal mapping — because the two indicator sets serve distinct purposes and use different characterisation factor packages.

Assertions added (19 EN 15804 individuals → 16 EF 3.1 individuals):

EN 15804 individual	→	EF 3.1 / LCIA individual
ind_GWPtotal	→	ind_EF_climateChange
ind_GWPfossil	→	ind_EF_climateChange
ind_GWPbiogenic	→	ind_EF_climateChange
ind_GWPluluc	→	ind_EF_climateChange
ind_ODP	→	ind_EF_ozoneDepletion
ind_AP	→	ind_EF_acidification
ind_EPfreshwater	→	ind_EF_eutrophicationFreshwater
ind_EPmarine	→	ind_EF_eutrophicationMarine
ind_EPterrestrial	→	ind_EF_eutrophicationTerrestrial
ind_POCP	→	ind_EF_photochemicalOzoneFormation
ind_ADPE	→	ind_EF_resourceUseMineralsMetals
ind_ADPF	→	ind_EF_resourceUseFossils
ind_WDP	→	ind_EF_waterUse
ind_PM	→	ind_EF_particulateMatter
ind_IRP	→	ind_EF_ionisingRadiationHumanHealth
ind_ETPfw	→	ind_EF_ecotoxicityFreshwater
ind_HTPc	→	ind_EF_humanToxicityCancer
ind_HTPnc	→	ind_EF_humanToxicityNonCancer
ind_SQP	→	ind_EF_landUse

6.2 `assessesImpactCategory`

Added between `LCIAImpactIndicator` and `LCIAImpactCategory`. Records the classification link between each EF 3.1 impact indicator individual and the mid-point impact category it characterises. Domain is restricted to `LCIAImpactIndicator` only — not extended to `EN15804ImpactIndicator` — because EN 15804 impact indicators are already linked to their EF 3.1 counterparts via `relatesToLCIAIndicator`, from which the impact category is reachable transitively. Adding a direct `assessesImpactCategory` link from EN 15804 indicators would be redundant.

Assertions added (16 EF 3.1 individuals → 16 impact category individuals):

EF 3.1 individual	→	Impact category individual
ind_EF_climateChange	→	climateChange
ind_EF_acidification	→	acidification
ind_EF_eutrophicationFreshwater	→	eutrophication
ind_EF_eutrophicationMarine	→	eutrophication
ind_EF_eutrophicationTerrestrial	→	eutrophication
ind_EF_photochemicalOzoneFormation	→	photochemicalOzoneFormation
ind_EF_ozoneDepletion	→	ozoneDepletion
ind_EF_ionisingRadiationHumanHealth	→	ionisingRadiationHumanHealth
ind_EF_resourceUseFossils	→	resourceUseFossil
ind_EF_ecotoxicityFreshwater	→	freshwaterEcotoxicity
ind_EF_humanToxicityCancer	→	humanToxicity
ind_EF_humanToxicityNonCancer	→	humanToxicity
ind_EF_landUse	→	landUseSoilQuality
ind_EF_resourceUseMineralsMetals	→	resourceUseMineralsMetals
ind_EF_waterUse	→	waterUseDeprivation
ind_EF_particulateMatter	→	respiratoryInorganics

6.3 `addressesAoP` — Individual Assertions

Individual assertions added between `LCIAImpactCategory` individuals and `AreaOfProtection` individuals, based on EF 3.1 (Andreasi Bassi et al., JRC, 2023):

Impact category individual	→	Area(s) of protection
climateChange	→	aopHumanHealth, aopNaturalEnvironment
ozoneDepletion	→	aopHumanHealth, aopNaturalEnvironment
humanToxicity	→	aopHumanHealth
respiratoryInorganics	→	aopHumanHealth
ionisingRadiationHumanHealth	→	aopHumanHealth
photochemicalOzoneFormation	→	aopHumanHealth, aopNaturalEnvironment
acidification	→	aopNaturalEnvironment
eutrophication	→	aopNaturalEnvironment

Impact category individual	→	Area(s) of protection
freshwaterEcotoxicity	→	aopNaturalEnvironment
landUseSoilQuality	→	aopNaturalEnvironment, aopNaturalResources
waterUseDeprivation	→	aopHumanHealth, aopNaturalEnvironment, aopNaturalResources
resourceUseFossil	→	aopNaturalResources
resourceUseMineralsMetals	→	aopNaturalResources
terrestrialEcotoxicity	→	aopNaturalEnvironment
marineEcotoxicity	→	aopNaturalEnvironment
ionisingRadiationEcosystems	→	aopNaturalEnvironment

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7. Major data Property Changes

7.1 `amount` — Domain Extension

Domain widened from `LCAModuleValue` alone to `LCAModuleValue ∪ LCAResult` (union domain). This allows an `LCAResult` instance to carry a direct numeric value for aggregated or non-module-specific declarations, while `LCAModuleValue` continues to carry module-specific values (e.g. per lifecycle module A1–A3, C3, D). The two usages are not mutually exclusive: a result can have both a direct `amount` and one or more `LCAModuleValue` instances with their own `amount`.

Updated `rdfs:comment`:

> *"Numerical value of an LCA result. When declared on an LCAModuleValue instance, it represents the value for a specific life cycle module (e.g. A1–A3, C3, D) in the unit given by the associated LCIA method. When declared directly on an LCAResult instance, it represents an aggregated or non-module-specific value."*

7.2 Standard Vocabulary Alignments (subPropertyOf)

The following data properties are declared as subproperties of standard Dublin Core terms:

- `dataSetVersion`, `EPDFormatVersion`, `ILCDFormatVersion`, `methodVersion`, `methodologyVersion`, `modelVersion` → `rdfs:subPropertyOf dcterms:hasVersion`
- `registrationNumber`, `UUID` → `rdfs:subPropertyOf dcterms:identifier`

7.3 Replaced by Standard Vocabulary

The following local data properties have been deleted and replaced by direct use of standard terms:

- `impactIndicatorDescription` → `dcterms:description`
- `baseName` → `rdfs:label`
- `modelName` → `rdfs:label`

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8. Major annotation Property Change

8.1 `ilcdSource` → `sourceVocabulary`

The annotation property `ilcdSource` has been renamed to `sourceVocabulary`. The original name framed the ontology as derivative of the ILCD data format, implying that ILCD was the reference point

against which all elements were measured. This framing is incorrect: this ontology is an LCA and EPD ontology designed for DPP purposes, which aligns with external vocabularies and standards — including ILCD — but does not originate from them.

Updated values:

Value	Status	Meaning
"LCA ontology"	New (merges previous two)	Native element; no direct external counterpart. Replaces the previous values "LCA ontology extension" and "LCA ontology metadata", which have been merged.
"ILCD base"	Retained	Originates from the ILCD 1.1 base data format
"ILCD+EPD extension"	Retained	Originates from the ILCD+EPD v1.3 extension
"Dublin Core"	New	Aligned with Dublin Core Terms
"PROV-O"	New	Aligned with W3C PROV-O
"SKOS"	New	Aligned with W3C SKOS
"FOAF"	New	Aligned with FOAF vocabulary
"LCAnet"	New	Aligned with LCAnet ontology

All `rdfs:comment` values and `sourceVocabulary` annotations across the ontology have been updated accordingly.

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9. Major named Individual Changes

All EN 15804 lifecycle module individuals shared IRIs with COMP conformity assessment module individuals (`moduleA1`, …, `moduleD`), causing a one-IRI–two-types conflict when both modules are loaded in the same graph. These have been renamed with the `en15804` prefix.

Other individuals of ImpactCategory also shared the same IRI with individuals of LCAIndicators. These were corrected.

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10. Major Connector and P_DPP Module Actions

The following cross-module declarations are **not part of this LCA module file** and must be added in the Connector module:

1. `LCAResult rdfs:subClassOf QuantitativeProperty` (and `LCAModuleValue rdfs:subClassOf QuantitativeProperty`) — to allow LCA results to inherit `hasMeasurementUnit` from P_DPP.

   Note that the LCAResult class has the following subclasses:



Accordingly, they will all be subclasses of `QuantitativeProperty`.

2. `hasActor` object property (`LCAStudy → Actor`).

P_DPP action required (based on Tarmo's proposal because these elements now exist in LCA module)

1. `EnvironmentalFootprint` class and its subclasses (MaterialFootprint and CarbonFootprint) to be deleted from P_DPP.
2. `ResourceConsumption` class and its subclasses (LandUse, SustainableRenewalMaterialsUse, REcycledMaterialsUse, WaterConsumption, and EnergyConsumption) to be deleted from P_DPP